Development of a 6-DOF Aircraft Maneuver Control System: Adaptive PID Controller Design Using Fuzzy Logic to Optimize Aerobatic Maneuvers with Hardware-in-the-Loop Technology.

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Project Brief Introduction

This project aims to enhance aviation safety by focusing on improving the controller design for aerobatic maneuvers. Initially, a realistic model of flight dynamics was developed to obtain a nonlinear system. Subsequently, an adaptive control design using look-up tables is employed to achieve the safe and effective execution of aerobatic maneuvers.

Method

To enable the aircraft to perform inner loop maneuvers, lookup tables have been utilized to calculate drag, lift, and moment forces, allowing control over the aircraft's maneuvers. Additionally, environmental effects have been incorporated into the flight dynamics model. The flight model is designed by taking into account changes in air density based on the aircraft's altitude, as well as environmental factors such as gravity and wind. Outputs such as roll, pitch, yaw, and aircraft speed, obtained using the 6-DOF block, have been communicated to the Flightgear application. This visualization enables the depiction of the aircraft's movements.

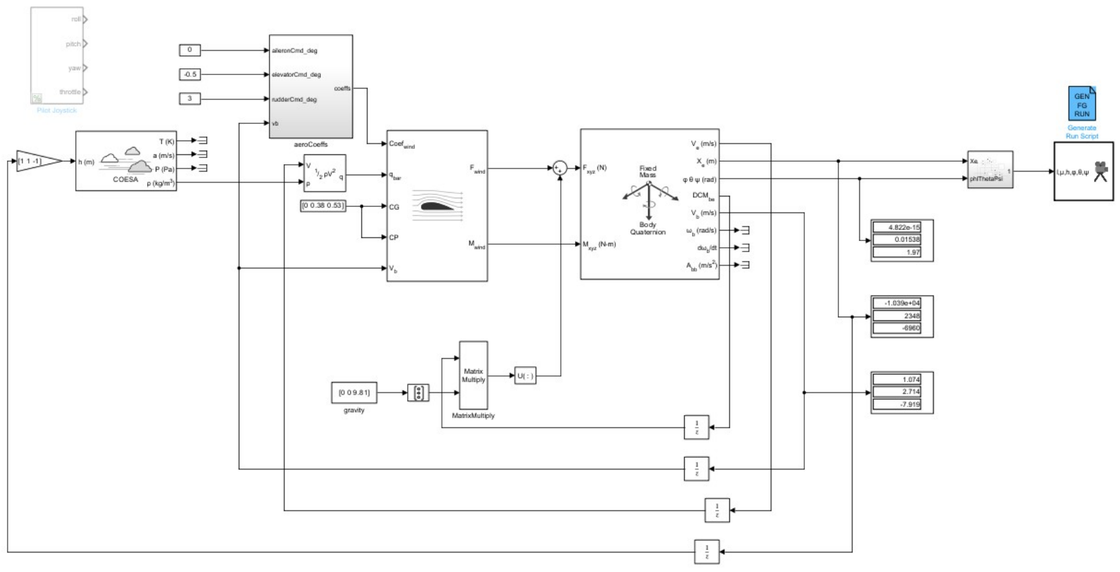


Figure 1 Simulink System Flight Dynamics Model

The input to the Simulink model given in Figure 1 consists of control surface angles such as aileron, elevator, and rudder, provided in radians. Based on these inputs, the aircraft's speed, position, roll, pitch, yaw angles, angular velocities, and angle of attack are calculated to determine and visualize the aircraft movement.

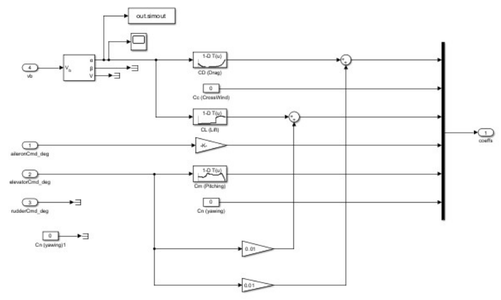


Figure 2 Simulink model where drag, lift, and moment force coefficients are determined based on look-up tables

In Figure 2, the angle of attack is calculated using the aircraft's speed in the body frame. The analysis of the wing profile is conducted, and appropriate drag, lift, and moment coefficients are obtained. Using lookup tables with the established angle of attack-coefficient relationship, the suitable coefficients are selected based on the angle of attack, and then drag, lift, and moment forces are calculated.

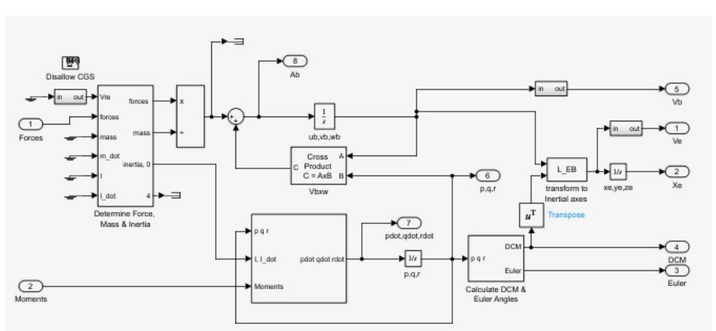


Figure 3 6-DOF Block

Project Plan

A timeline and workflow diagram outlining the tasks to be undertaken within the scope of the project.

Work Package	October	November	December	January
Literature Review				
Modeling of the Flight System				
Determination of Limits for Aerobatic Maneuvars				
Simulation Model Development				
Integrated studies and literature review for the Adaptive PID Controller Design with Fuzzy Logic Theory for the model.				
Research and literature review on the conversion of the model and controller into code, loading onto the microcontroller, and visualization of the project (GUI Design)				
Preliminary work for the graduation project				

Conclusion

The elevator angle is provided as an input for the aircraft. The motion of the elevator control surface results in pitch motion (nose-up and nose-down) along the lateral axis. This continuous motion completes the aircraft's internal roll maneuver. During the internal roll maneuver, the aircraft aims to maintain a constant angular velocity and strives to keep the radius value constant at every point of the circle it forms. Additionally, when examining the angle of attack of the aircraft, it is evident that the aircraft undergoes a complete roll motion. Instances of sudden transition from 0 to 180 degrees in a positive direction to -180 degrees are observed in the outputs. It is noted that the angular velocity of the aircraft accelerates with decreasing acceleration, and after a certain period, it is determined to have a more constant angular velocity. Angular velocity and angle of attack graphs, along with both the position video in two axes and the video from the FlightGear simulation application, are presented in figures 4, 5, 6, and 7 as output.

Figure 4 Matlab Animation video QR

Figure 5 Flightgear video QR

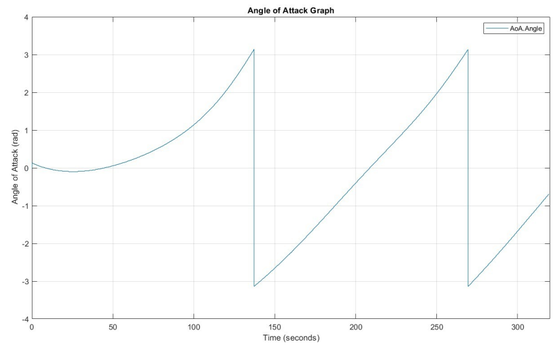


Figure 6 During the inner loop motion of the aircraft, a time-angle of attack angle graph.

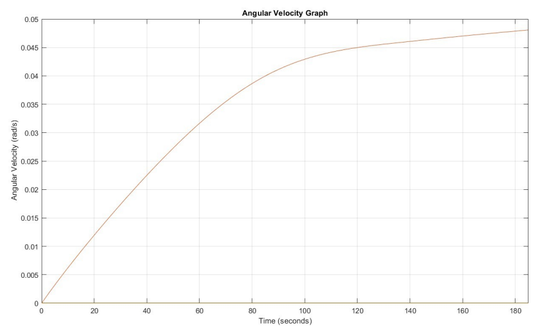


Figure 7 The angular velocity graph of the aircraft

Future Works and Plans

1.Designing a more robust controller for different acrobatic maneuvers
2.Integration of hardware in the loop test
3.To enhance the realism of the flight dynamics model by incorporating takeoff and landing dynamics